

by construction

$Sv = 0$ and $\ell c \le v \le uc$ hold exactly at v^*; the box is the GEM's, so it contains growth for every viable deletion

v^* enters the flux head detached, as a fixed feature of the current solve; only z^* carries a gradient

arm flux_off under the lp parameterization: no abundance head, no capacity ceiling, no thermodynamic term

measured (rounds 15b, 24 and 27)

round 15b: feasible from the first forward pass, the growth column ranks the held-out genes at 0.734 before training, and the forced cell receives no gradient when only exchange bounds bind; round 24: the flux head over this solve reads 0.852 on ProtT5; round 27: 0.882 on the pathway input, pooled 0.917